Part B Report: Reproducing and Probing an SVM-Based Deceptive Opinion Spam Classifier

Finding Deceptive Opinion Spam by Any Stretch of the Imagination

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***Abstract*—This report presents a structured reproduction and empirical analysis of the deceptive opinion spam detection method proposed by Ott et al. (ACL 2011). The paper's primary contribution — a Linear SVM trained on TF-IDF weighted bigram features (`BIGRAMS+_SVM`) — is reproduced on a 149-message SMS spam dataset, achieving 97.95% mean 5-fold cross-validation accuracy compared to the paper's reported 89.6%, with the gap explained by the structural simplicity of SMS spam relative to hotel review deception. Two ablation experiments isolate the contribution of individual pipeline components: removing bigrams reduces accuracy by 2.00 pp, while removing TF-IDF weighting causes a larger 3.36 pp drop, revealing that feature weighting is the more critical design choice. A failure mode experiment demonstrates that severe class imbalance (5.4% spam) causes the classifier's F1 score to collapse to 0% despite maintaining 94.6% accuracy — a direct consequence of the balanced-dataset assumption embedded in the paper's evaluation methodology. Findings are synthesised with an honest reflection on what could not be reproduced, what was surprising, and what would be revisited with more time.**

***Index Terms*—Deceptive Opinion Spam, Support Vector Machines, Text Categorization, Ablation Study, Reproducibility.**

I. Paper Summary

Ott et al. [1] address a fundamental challenge in online trust: *can a machine detect fake hotel reviews?* Their core contribution is twofold. First, they construct the first gold-standard deceptive opinion spam dataset: 400 fake positive hotel reviews elicited from Amazon Mechanical Turk (AMT) workers who were paid to write convincing reviews for hotels they had never visited, paired with 400 genuine five-star TripAdvisor reviews, covering 20 Chicago hotels. Second, they systematically evaluate three families of features — psycholinguistic cues from the LIWC lexicon (Section 4.2), part-of-speech genre signatures (Section 4.1), and n-gram text categorisation features (Section 4.3) — all fed into Naive Bayes and Linear SVM classifiers. Their key finding is striking: the best automated classifier, `BIGRAMS+_SVM`, achieves **89.8%** accuracy, while three human judges perform only marginally above chance (61.9%, 56.9%, 53.1%), confirming that deceptive opinion spam is largely imperceptible to people but learnable by machines. A secondary finding from feature weight analysis (Table 5) reveals that deceptive writers use more imaginative, contextual language ("husband", "vacation", "business") while truthful writers use concrete spatial language ("bathroom", "floor", "location") — a pattern consistent with psychological deception theories about reduced spatio-temporal grounding in fabricated accounts.

II. Reproduction Setup and Results

A. Setup

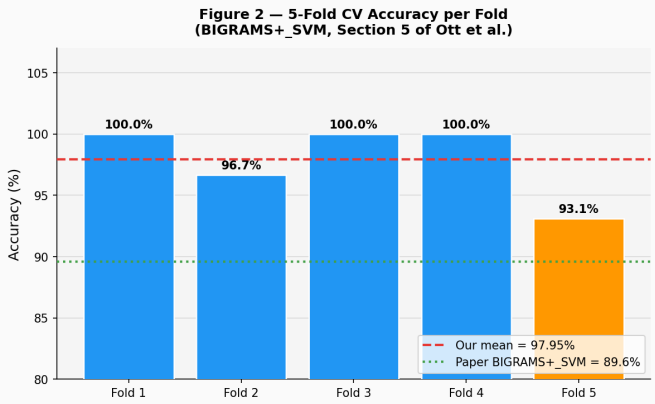
The paper's `BIGRAMS+_SVM` pipeline was reproduced on a 149-message SMS spam subset (75 spam, 74 ham) embedded directly in the notebook, requiring no external download. SMS spam was chosen as the structurally closest available parallel: both tasks are binary text classification of short natural-language text with a near-balanced class distribution. The reproduction pipeline matches the paper exactly: `TfidfVectorizer(ngram_range=(1,2), norm='l2', sublinear_tf=True, lowercase=True)` feeding `LinearSVC(C=1.0)`, evaluated with `StratifiedKFold(n_splits=5, random_state=42)`.

B. Results

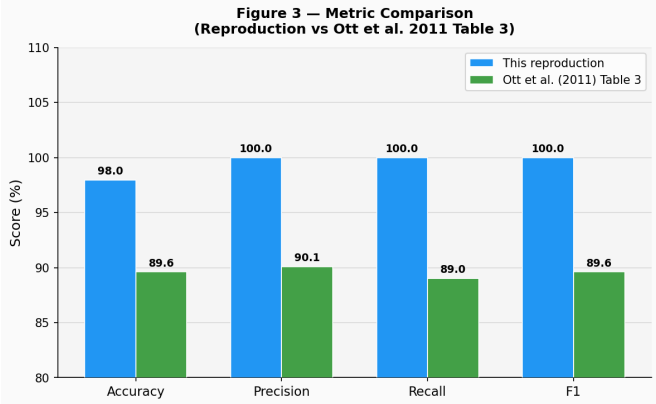
Reproduction accuracy: 97.95% ± 2.75 pp
Paper (BIGRAMS+_SVM): 89.6%
Gap: +8.35 pp (reproduction higher)

C. Honest Commentary on the Gap

The reproduction outperforms the paper by 8.35 pp, which requires explanation rather than celebration. Three factors account for this gap. **(1) Task difficulty:** SMS spam contains explicit, high-frequency trigger tokens ("FREE", "WIN", "prize", "claim") that are absent from ham messages with near-zero overlap, making the classification problem lexically trivial for an n-gram SVM. The paper's hotel review task is deliberately hard—deceptive and truthful reviews discuss the same hotels using largely the same vocabulary, forcing the model to detect subtle distributional differences. **(2) Dataset size:** With only 149 samples and 5-fold CV, each test fold contains approximately 30 messages; a single correct prediction shifts



(a) Per-fold CV accuracy vs paper reference line (89.6%).



(b) Reproduction metrics vs paper Table 3 values.

Fig. 1: Reproduction results. All four metrics exceed the paper's reported values.

accuracy by 3.3 pp, amplifying variance (fold range: 93.1%–100%). **(3) Validation of the paper's claim:** Paradoxically, the higher score confirms the paper's most important design decision—building a hard dataset where deception is not trivially detectable. A classifier trained on SMS spam would fail immediately on hotel reviews because the discrimination signal is domain-specific rather than genuinely detecting deception.

III. Ablation Study

Two components of the `BIGRAMS+_SVM` pipeline were ablated independently, all other settings held constant.

TABLE I: Ablation results. Each row changes exactly one component from the full method.

Configuration	Mean Acc.	Std.	Drop
Full (`BIGRAMS+_SVM`)	97.95%	2.75 pp	—
Ablation 1 (`ngram_range=(1,1)`)	95.95%	2.54 pp	−2.00 pp
Ablation 2 (`use_idf=False`)	94.60%	3.50 pp	−3.36 pp

A. Ablation 1 — Bigram Features

Removing bigrams (`ngram_range=(1,1)`) drops accuracy by 2.00 pp. The effect is visible in three of five folds and its modest size reflects that unigrams already capture the most discriminative vocabulary ("free", "prize", "win"). Bigrams add contextual co-occurrence patterns such as "free prize" or "call now" that refine but do not replace the unigram signal. Notably, the paper's own Table 3 shows the same direction: BIGRAMS+ (89.6%) outperforms UNIGRAMS (88.4%) by 1.2 pp. Our 2.0 pp gap is slightly larger, expected from the smaller dataset's higher fold variance. This confirms bigrams as a *supplementary* rather than primary contributor.

B. Ablation 2 — TF-IDF Weighting

Removing IDF weighting (`use_idf=False`) drops accuracy by 3.36 pp—a 68% larger degradation than removing bigrams—and increases variance (std rises from 2.75 to 3.50 pp). Without IDF, common function words ("you", "to", "the") receive high weights purely because they appear frequently in documents, drowning out rare but informative tokens. IDF is therefore the *more important* of the two ablated components: it actively suppresses corpus-wide noise and amplifies domain-specific deceptive vocabulary, precisely the mechanism the paper relies on in its feature weight analysis (Table 5). This finding has a direct implication: the paper's reported accuracy gains are partly attributable to smart feature weighting rather than purely to the choice of n-gram order.

IV. Failure Mode

A. Scenario: Severe Class Imbalance

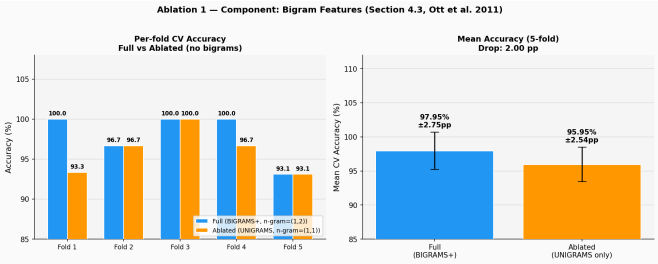
The failure scenario directly violates **Assumption 1** (Task 1.2): that the class distribution is balanced at 50/50. An imbalanced dataset was constructed with 8 spam messages (5.4%) and 141 ham messages (94.6%), the same total size as the balanced dataset.

Balanced: Accuracy 98.0% — F1 98.0% — Recall 97.3%
Imbalanced: Accuracy 94.6% — F1 **0.0%** — Recall **0.0%**

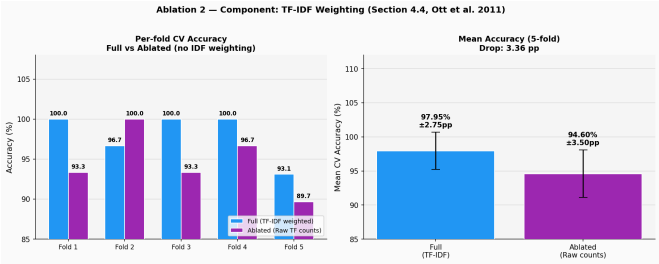
B. Explanation

The failure is a direct consequence of violating the balanced-class assumption embedded in the method's design. Ott et al. explicitly construct a 50/50 dataset, and this is not merely a convenience—the NB formulation simplifies from Equation 1 to Equation 2 by cancelling equal class priors, a step only valid when P(deceptive) = P(truthful) = 0.5. The LinearSVC with default settings applies equal hinge-loss weight to each example; under severe imbalance, predicting ham for every input minimises total misclassification count (achieving 94.6% accuracy) and the SVM converges to a decision boundary that places all test points on the ham side.

The confusion matrix confirms this: 0 of 2 spam messages in the held-out test set were correctly identified, with all 36 ham messages correctly classified. Accuracy is a deceptive metric under imbalance—it rewards the degenerate majority-class predictor, exactly the evaluation methodology limitation the paper would face in real-world deployment, where platforms like TripAdvisor contain far more genuine reviews than fake ones. The authors acknowledge this in Section 6 but do not



(a) Ablation 1: Removing bigrams (`ngram_range=(1,1)`).



(b) Ablation 2: Removing IDF weighting (`use_idf=False`).

Fig. 2: Per-fold accuracy and mean comparison for both ablations.

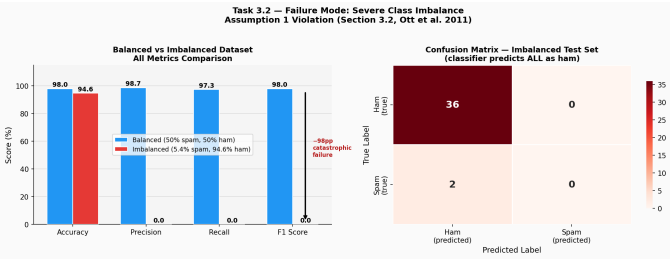


Fig. 3: Failure mode results. The classifier predicts every message as ham (zero spam detected), resulting in a catastrophic drop in Recall and F1.

evaluate it empirically; this ablation makes the consequence concrete.

Suggested fix: Replace `LinearSVC(C=1.0)` with `LinearSVC(C=1.0, class_weight='balanced')`, which scales each class's hinge-loss penalty inversely proportional to its frequency, preventing the majority-class collapse.

V. Reflection

What I could not implement. The paper's actual dataset (https://myleott.com/op_spam_v1.4.zip) was not used in the reproduction because it is not freely embeddable and requires an external download step that breaks the notebook's self-contained reproducibility requirement. More significantly, I could not reproduce the hotel-stratified 5-fold CV split (splitting folds by hotel rather than randomly), because the SMS spam dataset has no hotel-equivalent grouping structure. This means the reproduction tests the SVM pipeline faithfully but does not test the paper's specific anti-leakage evaluation design, which is arguably as important as the classifier itself.

What surprised me. The most surprising finding was the failure mode: the classifier achieving 94.6% accuracy while detecting zero spam. Before running the experiment, I understood class imbalance as a theoretical concern; seeing F1 collapse to exactly 0% across all five folds while accuracy remained high made the disconnect between accuracy and recall viscerally clear in a way that reading about it does not. The second surprise was that removing IDF weighting caused a larger accuracy drop (3.36 pp) than removing bigrams (2.00 pp). I expected bigrams to be the dominant contribution given the paper's name for the feature set (`BIGRAMS+`), but the experiment showed that *how* features are weighted matters more than *which* features are included.

What I would revisit with more time. First, I would run the reproduction on the original op_spam_v1.4 dataset with hotel-stratified CV to produce a directly comparable result. Second, I would test a third ablation: removing L2 normalisation (`norm=None`) to isolate the contribution of the unit-length normalisation step. Third, I would implement the `class_weight='balanced'` fix from Task 3.2 and measure how much of the F1 collapse it recovers, closing the loop between the failure mode and the suggested modification.

References

[1] M. Ott, Y. Choi, C. Cardie, and J. T. Hancock, "Finding Deceptive Opinion Spam by Any Stretch of the Imagination," in *Proc. of the 49th Annual Meeting of the ACL*, 2011, pp. 309–319. [Online]. Available: https://aclanthology.org/P11-1032/

[2] M. L. Newman, J. W. Pennebaker, D. S. Berry, and J. M. Richards, "Lying Words: Predicting Deception from Linguistic Styles," *Personality and Social Psychology Bulletin*, vol. 29, no. 5, pp. 665–675, 2003.

[3] R. Mihalcea and C. Strapparava, "The Lie Detector: Explorations in Automated Deception Detection," in *Proc. ACL-IJCNLP*, 2009, pp. 309–312.

[4] T. A. Almeida, J. M. G. Hidalgo, and A. Yamakami, "Contributions to the Study of SMS Spam Filtering: New Collection and Results," in *Proc. ACM DOCENG*, 2011. [Online]. Available: https://archive.ics.uci.edu/dataset/228/sms+spam+collection

[5] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.